

# Supply Chain Efficiency & Sales Forecasting

Understanding delivery performance, sales trends, and uncovering data-driven growth opportunities across a global supply chain network



# Critical Questions Driving This Analysis

## **Delivery Reliability**

Are we consistently meeting customer expectations for on-time delivery?

## **Delay Root Causes**

What factors are contributing most significantly to shipment delays?

## **Sales Performance**

How are sales trending across time periods and geographical markets?

## **Hidden Opportunities**

Are we overlooking profitable markets with untapped potential?



# Our Analytical Approach



## Data Preparation

Cleaned and standardised 180,000+ order records across multiple systems



## Exploratory Analysis

Identified patterns in delivery performance and sales behaviour



## Predictive Modelling

Built forecasts to anticipate future demand patterns



## Insight Delivery

Created interactive dashboards for ongoing business monitoring

*Tools: Google Sheets, SQL (BigQuery), Python (Prophet), Tableau*

# Delivery Reliability



Delivery Status Distribution by Shipping Mode

Late delivery

57.29%

Advance shipping

24.07%

Shipping on time

18.64%

Most Late Orders Are Delayed by Only 1–2 Days

80K

60K

40K

20K

0K

1

2

3

4

5

60,647

28,718

7,052

6,983

Shipping Mode

First Class

Same Day

Second Class

Standard Class

Order Region

Canada

Caribbean

Central Africa

Central America

Central Asia

East Africa

East of USA

Eastern Asia

Eastern Europe

North Africa

Northern Europe

Oceania

South America

South Asia

South of USA

Southeast Asia

Schedule Adherence Varies Significantly Across Shipping Mode

Shipping Mode

First Class

Same Day

Second Class

Standard Class

5.39%

59.69%

Although over half of orders are delivered late, most delays are limited to 1–2 days, indicating widespread but low-severity schedule slippage driven by shipping mode differences.

# Delivery Performance Reveals Critical Gaps

**57%**

## Late Deliveries

Over half of all orders failed to meet delivery commitments

**3-4**

## Days Delayed

Average delay for Second Class shipping method

- ❏ Shipping mode drives delays more than geography — indicating carrier-level performance issues rather than distance challenges

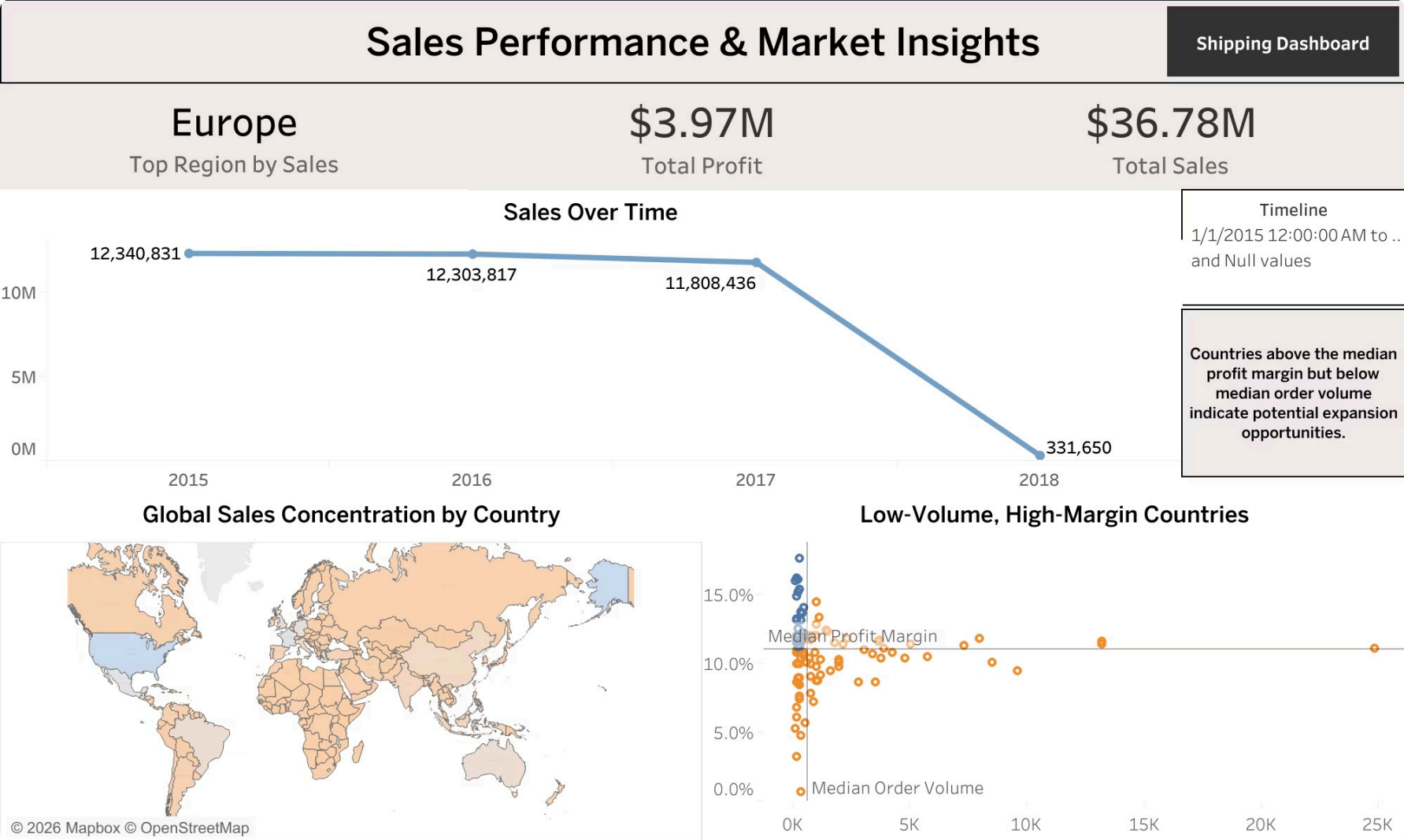


## What This Means

- Significant SLA performance gaps across shipping partners
- Customer satisfaction at risk with current delivery standards
- Opportunity to renegotiate carrier agreements and optimise shipping selections



# Sales Insights



Interactive dashboards analyzed delivery performance, delay patterns, and market profitability. These views identified systemic shipping delays, regional sales concentration, and high-margin market opportunities.



# Translating Insights into Business Impact

1

## **Optimise Shipping Operations**

Renegotiate carrier SLAs and prioritise faster, more reliable shipping modes to improve customer satisfaction

2

## **Target High-Margin Markets**

Direct sales and marketing resources towards low-volume, high-profitability regions for expansion

3

## **Align Operations with Demand**

Use forecasts to synchronise inventory levels, staffing, and logistics with anticipated seasonal peaks

These actions translate data findings into measurable improvements in customer experience, revenue growth, and operational efficiency

# Predictive Sales Modelling Enables Proactive Planning



01

## 12-Month Sales Forecast

Built predictive model using historical sales patterns and seasonal trends

02

## Q4 Demand Peaks Identified

Forecasts reveal significant seasonal surges requiring advance preparation

03

## Shift from Reactive to Proactive

Enables inventory planning, workforce allocation, and operational readiness ahead of demand



# Key Takeaways



## **Shipping Mode Matters Most**

Carrier selection and shipping class are the primary drivers of delivery delays, not distance or geography



## **Profitable Markets Exist Beyond Top Revenue**

High-margin opportunities hidden in markets with lower order volumes but stronger profitability per transaction



## **Forecasting Supports Better Decisions**

Predictive models enable proactive planning rather than reactive scrambling during demand surges

# Recommendations for Continued Growth



## Build Dimensional Data Models

Create scalable data infrastructure for deeper analysis and faster insights



## Automate Data Pipelines

Implement automated refresh cycles for dashboards and reports to maintain real-time visibility



## Enhance Forecasting Models

Incorporate external signals like market trends, economic indicators, and promotional calendars